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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-SE-004

1. Topic (12 words max)

Adaptive Cognitive Load Prediction System For Online Learning Using Behavioral Patterns

2. Research group the project belongs to

CEAI - Centre of Excellence in AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Online learning is widely adopted in higher education and professional training, yet many Learning Management Systems (LMS) still deliver static instruction uniform content, pacing, and navigation without responding to learners' moment-to-moment cognitive capacity and interaction behavior. This creates a persistent risk of cognitive overload (too much information/effort) or underload (low challenge), which can reduce comprehension, engagement, and retention and increase frustration and dropout risk [1]–[3]. In parallel, cognitive ergonomics problems in LMS interfaces (e.g., complex navigation, repeated actions, error-prone workflows) can increase mental effort and task difficulty, further degrading learning efficiency [4]. Evidence also suggests that behavioral/affective responses may mediate learning outcomes, and high visible engagement does not always mean effective learning—sometimes it reflects increased extraneous load highlighting the need for cognitive-state awareness rather than static presentation [5].

A core research gap is that most cognitive-load measurement approaches depend on intrusive or impractical methods such as eye tracking, EEG, or post-task questionnaires, which do not scale for real-world online platforms and cannot support continuous, real-time adaptation [1], [5]. In addition, learner cognitive styles (e.g., visual/verbal, reflective/impulsive) are commonly captured through self-report questionnaires and treated as stable, although they may shift across tasks and over time [2], [3]. Existing adaptive systems typically adjust difficulty or sequencing but rarely transform the same theory content into alternative representations matched to the learner's predicted load and inferred style [2], [5]. Finally, many AI-driven adaptations operate as black boxes, limiting educator trust and robustness under behavioral drift and large-scale deployment [4], [5].

Therefore, there is a need for a non-intrusive, behavior-driven, explainable system that (i) predicts cognitive load in real time from fine-grained interaction patterns, (ii) dynamically infers cognitive

style from observed behaviors, and (iii) delivers theory-level adaptive explanations while addressing core usability/ergonomics issues in LMS workflows.

References (IEEE)

[1] O. Mwambe and E. Kamioka, "Utilization of learners' metacognitive experiences to monitor learners' cognition states in e-learning platforms," International Journal of Information and Education Technology, vol. 9, no. 5, pp. 362–365, 2019.

[2] J. Khlaisang and N. Songkram, "E-Learning system to enhance cognitive skills for learners in higher education: A review of possibly open education," International Journal of the Computer, the Internet and Management, vol. 21, no. 1, pp. 17–23, 2013.

[3] E. Lavrov, V. Logvinenko, V. Osadchyi, O. Siryk, and Y. Chybiriak, "Ensuring the effectiveness of e-learning based on online technology: Analysis of factors influencing the cognitive independence of students," in Proc. SCITEPRESS, 2022, doi: 10.5220/0011009500003364.

[4] H. C. Ouertani and G. Al-Hudhud, "Optimizing e-learning cognitive ergonomics based on structural analysis of dynamic responses," International Journal of Emerging Technologies in Learning (iJET), vol. 14, no. 10, pp. 150–160, 2019, doi: 10.3991/ijet.v14i10.10134.

[5] Y. Sun and D. A. Nembhard, "Static vs. dynamic representations and the mediating role of behavioral affect on e-learning outcomes," International Journal of Human–Computer Interaction, 2022, doi: 10.1080/10447318.2022.2096187.

6. Brief description of Existing Research and Systems (300 words max)

AI-based Educational Platforms

AI-based educational platforms have increasingly been used to personalize learning, provide adaptive feedback, and improve student engagement across higher education and professional training. These systems leverage artificial intelligence to monitor learner performance, adjust content delivery, and facilitate mastery learning. However, while effective in specific domains, current platforms exhibit limitations in scalability, cognitive personalization, and dynamic content adaptation.

1) Carnegie Learning's MATHia

MATHia is an AI-driven cognitive tutoring system, primarily focused on high school algebra. It provides personalized practice, adaptive feedback, and mastery learning by tracking student progress and adjusting problem sequences [1].

2) Knewton Alta

Knewton Alta is an AI-powered adaptive learning system for higher education. It offers personalized content sequencing, mastery learning, and real-time assessment based on student performance [2].

3) Oxford Medical Simulation (Virtual Patient Training)

This platform uses AI-powered virtual patients to train medical students in clinical reasoning. Students practice decision-making in realistic scenarios while the AI monitors their actions and provides targeted feedback [3].

Gaps Across Existing Systems

- **Not using multiple AI agents:** MATHia, Knewton Alta, and Oxford Medical Simulation are all single-agent systems, limiting collaborative adaptation and integration of multiple learning signals.

- **Content not dynamically adjusted:** MATHia and Oxford Medical Simulation have mostly static learning materials; adaptation is largely restricted to problem difficulty, sequence, or scenario selection.
- **Cognitive styles not dynamically inferred:** Knewton Alta and Oxford Medical Simulation do not adjust content presentation based on evolving learner preferences or behavioral patterns.
- **Limited use of behavioral patterns:** Knewton Alta and Oxford Medical Simulation rely primarily on correctness and pre-defined metrics; fine-grained interaction data (clicks, scrolling, pausing, time-on-task) is rarely used for real-time adaptation.
- **Rigid learning pathways:** MATHia has predefined sequences, limiting flexibility in lesson order beyond mastery criteria.
- **Limited real-time adaptation:** Oxford Medical Simulation provides feedback that is often post-event or scenario-specific rather than continuous, restricting responsiveness to learner behavior.

References

[1] F. E. Fancsali, M. Sandbothe, and S. Ritter, "Orchestrating classrooms and tutoring with Carnegie Learning's MATHia and LiveLab," in *Proc. Workshop on Towards the Future of AI Augmented Human Tutoring in Math Learning*, co-located with the 24th Int. Conf. on Artificial Intelligence in Education (AIED 2023), Tokyo, Japan, 2023, vol. 3491, CEUR Workshop Proc., pp. 1–11. [Online]. Available: https://ceur-ws.org/Vol-3491/paper1.pdf

[2] A. Jones and I. Bomash, "Validating mastery learning: Assessing the impact of adaptive learning objective mastery in Knewton Alta," in *Artificial Intelligence in Education*, Lecture Notes in Computer Science, vol. 10947, J. Houben, Ed. Springer, 2018, pp. 433–437, doi: 10.1007/978-3-319-93846-2_81.

[3] E. Lavrov, V. Logvinenko, V. Osadchyi, O. Siryk, and Y. Chybiriak, "Ensuring the effectiveness of e-learning based on online technology: Analysis of factors influencing the cognitive independence of students," in *Proc. 1st Symposium on Advances in Educational Technology (AET 2020)*, SCITEPRESS, 2022. [Online]. Available: https://doi.org/10.5220/0011009500003364

[4] G. Kostopoulos, V. Gkamas, M. Orfanidou, and S. Kotsiantis, "Agentic AI in education: State of the art and future directions," *IEEE Access*, Oct. 2025. [Online]. Available: https://doi.org/10.1109/ACCESS.2025.3620473

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

This research proposes a **non-intrusive, AI-driven Cognitive Load Prediction and Adaptive Learning System** that operates entirely on **fine-grained learner behavioral patterns** generated during online learning activities. The solution directly addresses the gaps identified in Section 6, where existing e-learning platforms are largely static, rely on intrusive sensing technologies, or use post-hoc cognitive assessments that cannot support real-time adaptation.

Addressing Identified Gaps

Current cognitive load measurement approaches depend heavily on **physiological sensors (EEG, eye tracking)** or **self-reported questionnaires**, which are impractical for large-scale online learning environments. In addition, cognitive styles are often treated as static learner attributes derived from questionnaires, while instructional content is rarely transformed at the theory level to match individual cognitive needs. The proposed system addresses these limitations by introducing a **behavior-only, real-time, and explainable adaptive framework** that continuously monitors, infers, and responds to learner cognitive states during learning.

System Perspective: Input – Process – Output

Input

The system captures **non-intrusive behavioral data** generated naturally within the learning platform, including:

- Clickstream events
- Scrolling and navigation patterns
- Video pausing, replaying, and skipping
- Typing cadence and response latency
- Time-on-task and interaction frequency
- Navigation entropy and content traversal paths

No physiological sensors or explicit questionnaires are required during learning.

Process

The processing layer consists of interconnected AI components operating in an event-driven manner:

1. **Behavioral Feature Extraction Module**
   Raw interaction events are transformed into temporal and statistical behavioral features.

2. **Cognitive Load Prediction Engine**
   Machine learning models estimate intrinsic, extraneous, and germane cognitive load levels continuously and in real time.

3. **Dynamic Cognitive Style Inference Module**
   Learner cognitive styles are inferred implicitly from behavioral adaptation patterns and updated dynamically as interaction behavior evolves.

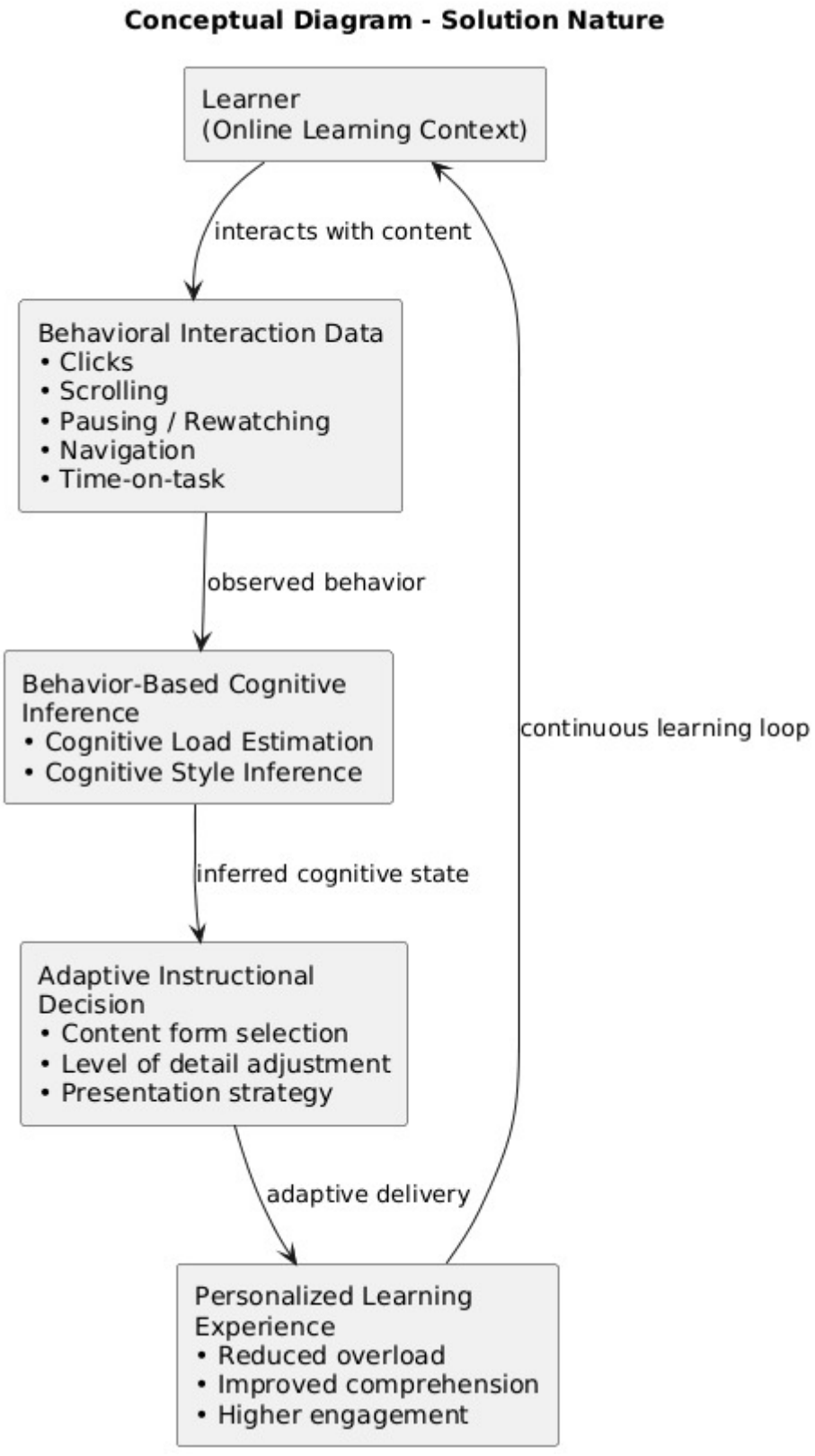
4. **AI-Based Content Transformation Agent**
   Core learning theory is automatically transformed into multiple representations (e.g., visual vs. textual, step-by-step vs. holistic) aligned with predicted cognitive load and inferred cognitive style.

5. **Explainability and Trust Layer**
   Explainable AI components generate human-interpretable reasons for adaptations, while drift detection mechanisms monitor behavioral and model changes over time to ensure reliability at scale.

Solution Nature Summary

Overall, the proposed solution represents a **behavior-driven, explainable, and scalable adaptive learning architecture** that enables real-time cognitive load prediction and theory-level content adaptation, making it suitable for deployment in real-world online learning platforms.

Conceptual Diagram



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The proposed research involves interdisciplinary expertise spanning educational psychology, cognitive science, artificial intelligence, machine learning, and software engineering. The project team will leverage existing knowledge in these areas and actively acquire specialized skills to ensure successful implementation.

Gaining Domain Expertise:
Team members can gain essential domain expertise in cognitive load theory, cognitive styles, and online learning pedagogy through a combination of academic resources and practical engagement. Key strategies include reviewing peer-reviewed literature on cognitive load, learning style models, and adaptive learning systems, as well as attending webinars, workshops, and short courses offered by recognized institutions in educational technology and cognitive science. Collaboration with faculty members or educational psychologists within the institution can provide guidance, mentorship, and access to domain-specific insights that enhance understanding of learning processes and behavioral indicators.

Acquiring Technical Knowledge and Skills:
The technical aspects of the project involve developing a microservices-based learning management system (LMS), event-driven architectures, multi-agent AI systems, and MLOps pipelines. Team members will build technical proficiency by undertaking hands-on training in modern tools and frameworks, such as Kafka or RabbitMQ for messaging, Docker and Kubernetes for orchestration, Python for AI/ML development, and cloud-based deployment strategies. Online courses, tutorials, and workshops in AI explainability, concept drift detection, and scalable system design will complement this skill development. Continuous peer learning within the team and iterative prototyping will also reinforce technical competencies.

Data Collection:
The data required for the project consists primarily of behavioral interaction logs from students using the learning modules, including click patterns, scrolling behavior, time spent on content, quiz performance, and navigation sequences. This data will be collected from voluntary participation in pilot studies conducted within the institution's LMS or simulated learning environments. Data anonymization and aggregation will ensure privacy while providing sufficient granularity for cognitive load prediction and cognitive style identification. Additionally, educators will contribute learning content for adaptation experiments, which will be processed by AI agents to create multiple representations aligned with learner profiles.

Ethical Considerations:
Given that the project involves human behavioral data, ethical clearance is required to ensure adherence to privacy, consent, and data protection standards. The research team will obtain Institutional Review Board (IRB) approval before data collection. Participants will provide informed consent, and all personal identifiers will be removed or pseudonymized. The system will comply with best practices for data minimization, secure storage, and controlled access to safeguard sensitive information. Furthermore, explainable AI mechanisms will be implemented to

provide transparency in content adaptation decisions, addressing ethical concerns around automated personalization.

In conclusion, the project team has a clear plan to acquire the specialized domain knowledge, technical skills, and ethical approvals necessary for successful completion. The availability of behavioral data, structured content from educators, and access to computational tools ensures that all aspects of the research can be effectively executed while maintaining compliance with ethical and professional standards.

9. Objectives and Novelty

Main Objective

To design, develop, and evaluate a scalable and intelligent online learning system that predicts students' cognitive load and cognitive styles using behavioral patterns, and dynamically adapts learning content through explainable, multi-agent AI while ensuring reliability, resilience, and long-term model accuracy through event-driven architecture and concept drift handling.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Subasinghe M.H.R.l. IT22089786	Real-time Cognitive Load Prediction	• Collect fine-grained learner behavioral data (clicks, scrolling, pauses, time-on-task) • Design short-term behavioral features using event streams• Build and train a real-time cognitive load prediction model • Validate prediction accuracy against learning performance indicators	Predicts cognitive load **non-intrusively** using real-time behavioral patterns without relying on questionnaires or physiological sensors, enabling continuous and scalable monitoring in online learning environments.

Imesh L.A.R IT22281364	Behavior-Based Cognitive Style Identification	• Aggregate long-term learner interaction data across sessions • Extract stable behavioral features representing learning preferences • Develop a machine learning model to infer cognitive styles dynamically• Continuously update learner profiles over time	Identifies cognitive styles implicitly and dynamically from learner behavior instead of static self-reported surveys, allowing cognitive styles to evolve with learning context.
Jayasinghe J M S N IT22166074	Cognitive Load and Style-Aware Content Adaptation	• Design an AI agent to process core theory content uploaded by educators • Generate multiple representations of the same theory (simplified, visual, example-based) • Match adapted content with predicted cognitive load and cognitive style• Evaluate learning outcomes and engagement	Introduces theory-level content transformation, where the same learning concept is automatically adapted to both cognitive load and cognitive style, rather than only adjusting difficulty or sequencing.

Dilshan W.G.W IT22055712	Design a scalable, resilient, explainable, and drift-aware multi-agent AI architecture for adaptive online learning systems	• Implement event-driven communication using message queues • Apply backpressure to control event flow under load • Use caching to reduce latency in AI-driven adaptation • Enable graceful degradation during service or model failures • Develop explainable multi-AI agents for prediction and recommendation • Detect concept drift in learner behavior over time • Mitigate drift via retraining and fallback strategies • Monitor system reliability and model performance	Introduces a drift-aware, resilient, and explainable multi-agent AI architecture that integrates message queues, caching, backpressure, graceful degradation, and concept drift handling to deliver trustworthy and scalable cognitive-aware personalization.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

— No —

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Kalpani	Manathunga	
Co-Supervisor	Ms	Jeenala	Perera	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				